

JOHN (CHANG WON) LEE

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EDUCATION

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| University of Toronto
PhD in Aerospace Engineering with a specialization in Robotics
• Advisor: Steven L. Waslander | Toronto, Canada
Sep 2025 – Nov 2029 (expected) |
| University of Toronto
MAsc. in Aerospace Engineering with a specialization in Robotics
• Advisor: Steven L. Waslander | Toronto, Canada
Sep 2023 – Sep 2025 |
| University of Toronto
BAsc. in Engineering Science (Machine Intelligence Major) | Toronto, Canada
Sep 2018 – Jun 2023 |

TEACHING

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|---|---------------------|
| • Teaching Assistant , Robotics Capstone Design (ROB498)
University of Toronto, Institute for Aerospace Studies | Jan 2026 – Apr 2026 |
| • Teaching Assistant , Mathematics for Robotics (ROB310)
University of Toronto, Institute for Aerospace Studies
• Taught in Fall 2023, Fall 2024, and Fall 2025. | Sep 2023 – Dec 2025 |
| • Teaching Assistant , Artificial Intelligence (ROB311)
University of Toronto, Institute for Aerospace Studies | Jan 2025 – Apr 2025 |
| • Teaching Assistant , Introduction to Mobile Robotics (CSC477)
University of Toronto, Department of Mathematical & Computational Sciences | Jan 2024 – Apr 2024 |
| • Teaching Assistant , Linear Algebra for Engineering (MAT188)
University of Toronto, Department of Mathematics | Sep 2022 – Dec 2022 |

SELECT PUBLICATIONS & SUBMITTED WORK

- **C. W. Lee**, M. Yuan, and S. L. Waslander, “DistillPlan: Distilling Episodes into Decomposition Schemas for Efficient Long-Horizon Embodied Task Planning,” under review.
- **C. W. Lee**, S. Leveugle, P. Grouchy, C. Langley, S. Stolpner, J. Kelly, and S. L. Waslander, “FlowCLAS: Enhancing Normalizing Flow-Based Anomaly Segmentation Via Contrastive Learning,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. [pdf]
- S. Leveugle, **C. W. Lee**, S. Stolpner, C. Langley, P. Grouchy, S. L. Waslander, and J. Kelly, “A Photorealistic Dataset and Vision-Based Algorithm for Anomaly Detection During Proximity Operations in Lunar Orbit,” *IEEE Robotics and Automation Letters*, vol. 11, no. 3, pp. 2418–2425, 2026. [pdf]
- **C. W. Lee** and S. L. Waslander, “UncertaintyTrack: Exploiting Detection and Localization Uncertainty in Multi-Object Tracking,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 4946–4953. [pdf]
- J. Deery, **C. W. Lee**, and S. L. Waslander, “ProPanDL: A Modular Architecture for Uncertainty-Aware Panoptic Segmentation,” in *2023 20th Conference on Robots and Vision (CRV)*, 2023, pp. 137–144. [pdf]

PROFESSIONAL & RESEARCH EXPERIENCE

Graduate Student Researcher

University of Toronto Robotics Institute

Sep 2023 – Present

Toronto, Canada

- Developed DistillPlan (under review), an embodied task planner that distills past episodes into parameterized decomposition schemas and calls them like functions. On ALFRED (LoTa-Bench protocol), it improves goal success by ~ 3 percentage points over ReAcTree while generating 78% fewer tokens.
- Lead a three-person dual-arm manipulation team using a Trossen Stationary AI and an InDro Axiom with Pika arms. Collect teleoperation data for imitation learning, fine-tune vision-language-action (VLA) policies ($\pi_{0.5}$, GR00T), and set up Isaac Sim / Isaac Lab for training.
- Build a system in which coding agents (Claude Code, Codex) write arm-control code through a custom robot harness to recover when a VLA policy fails. Explore online, continual VLA fine-tuning from these recoveries.
- Developed FlowCLAS, a normalizing flow-based anomaly segmentation method that adds contrastive learning and outlier exposure. Achieved state-of-the-art results on multiple benchmarks (WACV 2026).
- Built an automated Blender pipeline to generate a large-scale photorealistic lunar-orbit dataset, establishing a benchmark for visual anomaly detection and segmentation in space robotics (RA-L 2026).
- Designed UncertaintyTrack, a 2D multi-object tracker that uses learned localization uncertainty to stay robust under occlusion and non-linear motion (ICRA 2024).

Software Engineer Intern

16 Bit

Jan 2023 – Apr 2023

Toronto, Canada

- Built a PyTorch tool for bone mineral density (BMD) test analysis using object detection and OCR, cutting analysis of hundreds of daily cases from minutes to seconds.
- Deployed it with ONNX and Docker, integrating it into 16 Bit's existing medical imaging services.

Research Assistant

University of Toronto

Dec 2021 – Aug 2022

Toronto, Canada

- Implemented and benchmarked state-of-the-art detection and segmentation models in PyTorch, setting baselines for long-range traffic detection and panoptic segmentation research (CRV 2023).
- Built a MATLAB camera calibration pipeline and ran it on 100+ hours of traffic footage to support data annotation for autonomous-driving perception.

Software Engineer Intern

The Citco Group Limited

May 2021 – Apr 2022

Toronto, Canada

- Deployed a Dockerized search engine for financial application logs using the Elastic stack and custom regex patterns, improving log searchability by about 500% and reducing downtime.
- Classified and organized financial documents in PostgreSQL using character recognition libraries and a custom REST API, speeding up document lookups.
- Wrote automated JavaScript tests with Cypress for Citco's investment platform.
- Worked with cross-functional teams through daily scrums and other Agile practices.

ACTIVITIES & SOCIETIES

aUToronto Software Engineer

Aug 2021 – Apr 2023

- Built a 3D multi-object tracking system for autonomous driving in C++/Python using Extended Kalman Filters and Hidden Markov Models, improving accuracy by 10% and efficiency by 5%.
- Integrated the tracker into the car's ROS-based perception pipeline, contributing to the team's first-place finishes in multiple categories of the SAE AutoDrive Challenge II.

University of Toronto Engineering Student Ambassador

Aug 2019 – Aug 2022

- Represented Engineering Science at orientations, information sessions, and club fairs.

University of Toronto Engineering Student Mentor

Sep 2020 – Aug 2021

- Mentored first-year engineering students on academic success and career development, and connected them with university resources.

Engineering Science Education Conference (ESEC) Web Designer & Organizer Jun 2020 – Jan 2021

- Designed and built the ESEC 2021 conference website, with keynote profiles, event details, and preparation resources. [link]
- Worked with the organizing committee to bring in industry speakers who helped students choose their Engineering Science specialization.

HONORS, AWARDS & CERTIFICATES

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| • Dean's Honor List | 2018 – 2023 |
| • Natural Sciences and Engineering Research Council of Canada (NSERC)
Undergraduate Student Research Award (USRA) — \$7500 CAD | 2020, 2022 |
| • University of Toronto Scholar's Award — \$7500 CAD | 2018 |
| • Official Bilingual Certificate (English, French) | 2018 |

LANGUAGES

English (Native), **French** (Native), **Korean** (Native)

LINKS

- **Personal Website:** <https://johncwlee.github.io/>
- **Linkedin:** <https://www.linkedin.com/in/lee-cw-john/>
- **ESEC Website:** <https://engsci.utoronto.ca/experience/esec/esec-2021-home/>